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Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

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Ai miei meravigliosi nonni:
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Abstract

High temperature superconductivity in copper oxides (cuprates) represents one of the most striking phenomena in condensed matter physics. These materials host superconducting states with the highest known critical temperatures, emerging upon doping an antiferromagnetic insulator compound. The antiferromagnetic (AF) insulating and superconducting (SC) phases defy standard descriptions in terms of band picture and conventional phonon-mediated superconductivity. In contrast, these phenomena are genuine signatures of strong electronic correlations which hinder effective single-particle descriptions of the many-electron system.

After 40 years since their discovery, it is widely accepted that the Hubbard model on the square lattice provides a minimal description of the low-energy electronic properties of cuprates. The model describes the competition between the electronic kinetic energy, modelled by nearest-neighbours hopping t, and the Coulomb repulsion, modelled by a purely local interaction potential U. At half-filling, this competition results in the so-called "Mott localisation" phenomenon, which distinguishes the metallic phase for $U/t \ll 1$ from an insulating phase for $U/t \gg 1$. The latter is characterised by the almost complete localisation of electrons due to the high energetic cost of hopping towards occupied sites. Interestingly, this basic competition is intimately related to AF ordering and, upon doping, to the formation of the SC state with characteristic d-wave symmetry.

The fact that a purely repulsive interaction U is responsible for the formation of a SC state is remarkable, being superconductivity typically associated with an effective attractive interaction. In fact, a d-wave SC state can also be obtained, in general, by considering an attractive potential between electrons of neighbouring sites. Motivated by this observation, in this thesis, we studied a modified version of the Hubbard model, dubbed as the "extended Hubbard model", in which we include a nearest-neighbours attractive potential $-V$. The general goal of this work is to explore the interplay between the purely local repulsion and the non-local attraction. It is worth mentioning that this is not only motivated by the fact that both interactions are, in principle, responsible for d-wave superconductivity, but also by recent experimental observations which strongly suggest that the term $-V$ needs to be included in order to accurately describe the magnetic spectra of some specific compounds.

In this thesis, we study the extended Hubbard model using the Hartree-Fock method. Namely, we approximate the ground state by implementing a variational search in the subspace of the many-body Hilbert space of gaussian states. Despite its simplicity, the Hartree-Fock method provides a powerful framework to study competition among different phases, originating from different symmetry breaking mechanisms. In particular, we study the competition between three different phases: the normal phase, characterised by no symmetry breaking, the AF phase, characterised by a lowering of SU(2) symmetry, and the SC phase, characterised by the breaking of U(1) charge conservation.

This thesis work led to a number of original results, which can be summarised by the general observation that the non-local attraction $-V$ acts cooperatively with the local repulsion U for the stabilisation of the various phases hosted by the Hubbard model.

For the normal phase, we find that the non-local attraction acts as a source of Mott localisation, leading to an effective reduction of the electronic bandwidth. In particular, for values of V larger than a critical value and close to half-filling, we find a strong Mott localisation characterised by the complete flattening of the electronic bands. Interestingly, this result shows an example of Mott localisation that can be described at the level of the Hartree-Fock approximation, whereas Mott localisation in the conventional Hubbard model usually requires more sophisticated approximations.

For the AF phase, we restrict our analysis to the half-filled case. We find that the pairing-induced Mott localisation leads to a significant enhancement of the magnetisation, which primarily originates from the local repulsion U.

Finally, we focus on the superconducting phase and derive self-consistency equations in both singlet and triplet Cooper pairing channels, showing how Cooper pairing can be mediated simultaneously by both the local repulsion and the non-local attraction. Therefore, we numerically investigate the singlet channel and find evidence of "symmetry mixing", namely, the formation of a superconducting gap that is neither symmetric nor anti-symmetric under rotations of $\pi/2$ in the plane. We show that symmetry mixing is connected to the emergence of a finite nematic order parameter, that signals the breaking of planar rotational symmetry. This nematic symmetry breaking results in the fact that the renormalisation of bands becomes anisotropic, producing a difference in electronic kinetics in the x and y directions.

To summarise, the original results obtained in this thesis show that the non-local pairing provides a different route towards the formation of key strongly correlated phenomena, ranging from Mott localisation to nematic superconductivity, and call for future investigation using more sophisticated methods such as, for example, Dynamical Mean Field Theory.

Contents

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
DoF	Degree of freedom
DoS	Density of states
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PHS	Particle-hole symmetry
RMP(s)	Renormalized model parameter(s)
SC	Superconducting
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

Introduction

The discovery of superconductivity in copper oxide compounds, commonly referred to as cuprates, is one of the major breakthroughs in condensed matter physics in the last 50 years. In 1986, Bednorz and Muller [1] discovered that the compound BaLaCuO exhibits superconductivity below a critical temperature around 30K. Until then, superconductivity had been observed up to critical temperatures (T_{c}) of few Kelvins. This new class of superconductors marked the beginning of the era of high-T_{c} superconductivity. Today, the highest observed value is $T_{\mathrm{c}} \approx 151$ K at ambient pressure on the cuprate $\mathrm{HgBa_2Ca_2Cu_3O_{8+\delta}}$ [2].

Cuprates are the prototypical example of quantum materials characterised by rich phase diagrams, hosting very different phases upon changing external parameters. In Fig. 1.1 we sketch a typical example of the cuprates phase diagram as a function of temperature T and doping δ, with respect to the parent compound, at $\delta = 0$ [3]. The parent compounds are antiferromagnetic insulators below a Néel temperature $T_{\mathrm{N}} \approx 300$K. Upon doping with either electron or holes, the Néel temperature decreases, until antiferromagnetism is eventually suppressed. Increasing doping, the system undergoes a superconducting phase transition with a critical temperature T_{c} that shows a dome-like structure, and is maximised at an optimal doping δ^*. The optimal doping discriminates an underdoped region ($\delta < \delta^*$), where the system shows a bad metallic behaviour, and an overdoped region ($\delta > \delta^*$), where it becomes a conventional Fermi liquid. The full microscopic understanding of such complex phase diagrams and the interplay between the different phases have been subjects of an intense research activity in the last 40 years, and it remains, to date, an open question.

Today, it is generally accepted that the physics of cuprates is highly determined by strong electronic correlations, namely the fact that the electronic properties cannot be described by standard Fermi liquid theory. The simplest theoretical framework to describe strong electronic correlations in cuprates is the single-band Hubbard model which, as we are going to explain, is able to capture the formation of the antiferromagnetic state at zero doping, as well as the emergence of a d-wave superconducting state upon doping.

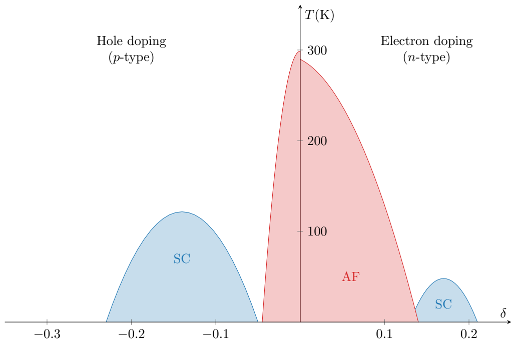

Figure 1.1 Sketch of the phase diagram of cuprates. These curves result from graphic elaboration and do not refer to a specific physical dataset.

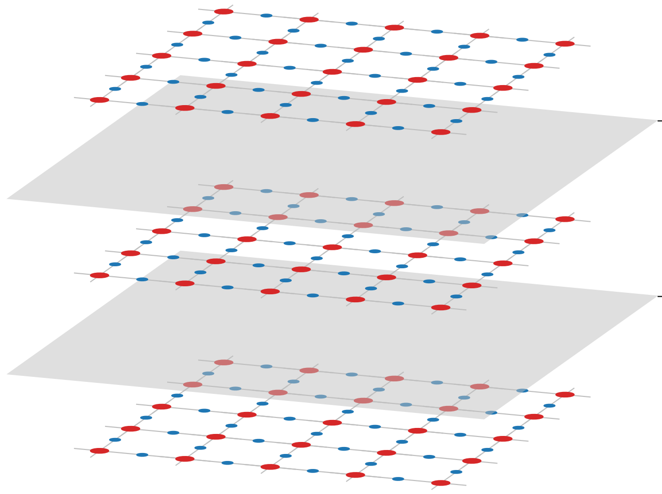

Figure 1.2 Schematics of a typical layered copper oxide compound. The inter layers are represented as solid planes, while the electrically active layers as a square grid hosting copper on its nodes and oxygen on its edges.

1.1 Cuprates and the square Hubbard model

The crystal structure of copper oxide superconductors is given by a stacking of 2D CuO_2 sheets separated by inert layers of insulating materials, see Fig. 1.2. The intermediate layers act as reservoirs of electrons (or holes) and dope the active CuO_2 sheets [4]. Due to the presence of these intermediate inert layers, the CuO_2 sheets are weakly coupled to each other, and can be treated as independent. In these sheets, copper and oxygens are arranged as a checkerboard lattice: oxygen atoms form a square lattice, and in the center of each cell is placed a copper atom. At zero doping, due to the high electronegativity of oxygens, copper atoms are ionised and left with one hole in their d shell, while oxygen atoms reach a closed shell configuration. Therefore, at low energy the kinetics of electrons is given by the hopping among the d orbitals of Cu^{2+} ions, arranged in a planar square lattice [5]. We notice that the number of electrons per unit cell is odd. Therefore, according to band theory, the system should be an half-filled metal, while instead it exhibits an insulating behaviour. Band theory relies on the assumption that electrons are nearly-free, and its failure in this context strongly suggests that the antiferromagnetic insulating phase originates from a strong Coulomb repulsion between electrons.

The simplest model capable of describing this situation is the single-band Hubbard model [6],

$$\hat{H}_{\mathrm{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad (t, U > 0), \tag{1.1}$$

being t the hopping energy and U the Coulomb repulsion, simplified to a local on-site interaction. Sum is intended on the spin index $\sigma \in \{\uparrow, \downarrow\}$ and the sites i, j of a planar square lattice $\mathcal{L}$. The notation $\langle ij \rangle$ indicates nearest neighbours (NNs).

The Hubbard model captures some of the low-energy properties of copper oxides [5, 7] and is capable of showing a rich variety of physical phenomena [8, 9]. Indeed, despite its simplicity, it can describe the Mott insulating state, antiferromagnetism and the formation of the d-wave superconducting state. These are three key phenomena for the the physics of cuprates.

1.1.1 The Mott insulating state in the Hubbard model

Conventional band theory describes the electronic properties of materials on the assumption that electron-electron interaction is negligible. Within this framework, a material forms a band insulator whenever the conduction band fills up. Due to Pauli principle, electrons fill all states of the band

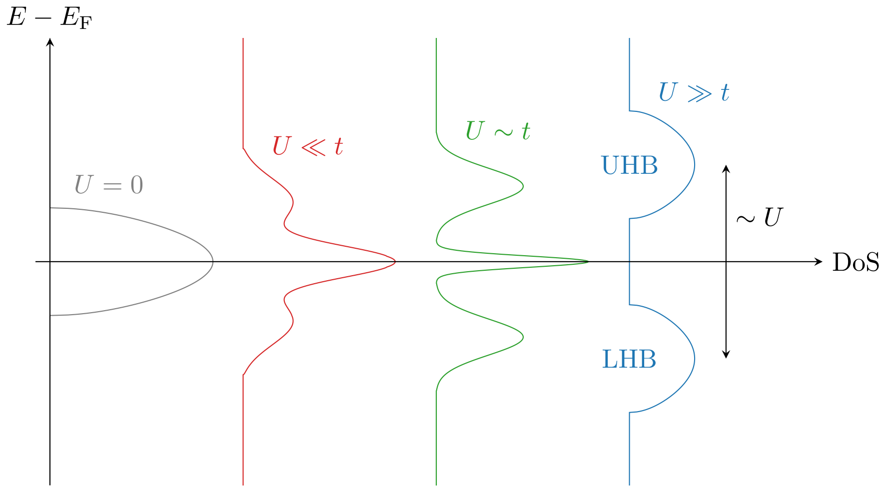

Figure 1.3 | Sketch of the Hubbard DoS as we vary the ratio U/t. For $U = 0$ (gray plot on the left) we have a free electronic band. As we increase U, two peaks appear while the central coherent peak shrinks. At $U \gg t$ (blue plot on the right), the DoS is separated in upper and lower Hubbard bands (UHB, LHB). E_{F} is the Fermi energy.

and have no room left for low-energy scattering, hence the filled band is completely ineffective for conduction. On the other hand, if the conduction band is partially filled, the material is a metal [4]. In general, band theory predicts a metallic behaviour if the number of electrons per unit cell is odd; an insulating behaviour if is even.

In the Mott insulating state, the insulating properties do not derive from a single-particle band being filled up and separated by a gap from another band at higher energy; instead, from the effect of the Coulomb repulsion that "freezes" electrons on the lattice sites [3]. One can picture this situation as a sort of traffic jam, where electrons do not move because the energetic cost of having two electrons on the same site would be too high [10]. This produces an insulating behaviour: electrons are completely localised and conductivity is low.

The Hubbard model describes naturally Mott physics [8]. Consider for instance the two-sites Hubbard model,

$$\hat{H} = -t\left(\hat{c}^{\dagger}_{1\uparrow}\hat{c}_{2\uparrow} + \hat{c}^{\dagger}_{2\downarrow}\hat{c}_{1\downarrow}\right) + U\left(\hat{n}_{1\uparrow}\hat{n}_{1\downarrow} + \hat{n}_{2\uparrow}\hat{n}_{2\downarrow}\right), \tag{1.2}$$

where the number subscript indicates the site index. In the strong coupling limit $U \gg t$, we can treat the kinetic terms perturbatively. Indeed, let E_i be the energy given by having i electrons on one site. At order zero in perturbation theory, the difference in energy between the state with one site doubly occupied and the other empty, and the state with two singly occupied sites is

$$E_2 + E_0 - 2E_1 = U.$$

The zeroth order ground-state, then, is a combination of states where both sites are singly occupied. Insertion of the kinetic terms as a perturbative correction does not change the scenario (see App. **??** for details).

In the Hubbard model, the transition from a metallic state to a Mott insulator is controlled by the strength of the Coulomb repulsion U, and can be described by the means of Dynamical Mean Field Theory [23]. The Mott transition impacts visibly the density of states (DoS), which undergoes significant changes [11], as we sketch in Fig. 1.3. Let us describe the general behaviour: to start, let $U = 0$ and consider a lattice with one electron per site (half-filling). Electrons are free and do not interact, therefore move freely across the lattice. Due to Pauli exclusion principle, they form a Fermi sea that occupies the lower half a metallic band with a width $W = \mathcal{O}(t)$ (gray plot on the left of Fig. 1.3). As we increase the value of U, it becomes increasingly harder for electrons to move. When U becomes sufficiently large (with respect to t), the electronic density of states splits in two bands (blue plot on the right of Fig. 1.3). The lower Hubbard band (LHB), filled, accommodates all completely localised electrons; in the upper Hubbard band (UHB), empty, are all the states where a site gets doubly occupied. This splitting is of order $\sim U$. At intermediate values of U (central plots of Fig. 1.3), the DoS presents a peculiar three-peaks shape, which is the result of a mixture of the two Mott bands, and a residual peak of the original free band around the Fermi energy E_F. One of the signatures of a Mott transition, thus, is the flattening of the bare single-particle bands.

1.1.2 Antiferromagnetism in the Hubbard model

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Indeed, in the strong coupling regime $U \gg t$, the Hubbard hamiltonian is effectively approximated by the Heisenberg hamiltonian with antiferromagnetic coupling [13],

$$\hat{H}_\mathrm{HM} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j,$$

where $\hat{\mathbf{S}}_i$ is the spin operator on site i.

Antiferromagnetism is established through the "super-exchange mechanism", which takes advantage of the fact that electrons are localised on-site. This is best explained on the two-sites Hubbard model toy model of Eq. (1.2): as we explained in Sec. 1.1.1, the kinetic term can be treated as a perturbation to the fully-localised ground-state, which favours the formation of a spin singlet between the two sites (see App. **??** for details). A spin singlet

The idea that singlet pairing across nearest-neighbouring sites could explain the insurgence of antiferromagnetism has been explored. A many-body state resulting from a superposition of spin singlets across different sites has been proposed [14, 15] to capture the relation between antiferromagnetism and Mott physics.

1.1.3 Unconventional superconductivity

Injection of charge (electrons or holes) in copper oxide compounds disrupts antiferromagnetic ordering and establishes a variety of peculiar phases [3]. Among these, the high temperature superconducting phase is significantly different from that described by Bardeen-Cooper-Shrieffer (BCS) theory. Because of this, is denominated "unconventional".

In BCS theory, couples of electrons are "glued" together by an effective attraction that is mediated by lattice phonons. This mechanism is referred to as "Cooper pairing". The resulting state is a condensate of Cooper pairs, with an energy spectrum that is fully gapped and thus no low-energy excitations. The gap $\Delta_\mathbf{k}$, as a function of momentum, is isotropic (s-wave) and represents the energy cost of breaking a Cooper pair glued by phonons.

In copper oxides, superconductivity is established with significantly different features. Pairing is still active, but is not mediated by phonons. Most notably, the superconducting gap is not isotropic but changes sign under rotations of $\pi/2$ on the plane [8]. Because of this, we talk about $d_{x^2-y^2}$-wave superconductivity. A consequence is the appearance of gapless quasiparticle excitations in the energy spectrum. Ideed, a $d_{x^2-y^2}$-wave gap has nodal lines (continuous sets of points, in reciprocal space, where it is null) which, in a square lattice geometry, result in a set of nodes around which dispersion is linearised (Dirac cones) [16]. Moreover, Mott physics and $d_{x^2-y^2}$-wave superconductivity are related. Indeed, such a spatial structure of the gap function is understood to be a manifestation of the fact that electrons are spatially correlated in such a way to avoid occupying the same site [8].

The source of pairing in cuprates is still object of debate. In the phase diagram of copper oxides, the $d_{x^2-y^2}$-wave superconducting region "emerges" from the antiferromagnetic region, as doping is increased at low temperature. Hence, it has been proposed that the same Mott-induced spin fluctuations that lead to the formation of the antiferromagnetic state could act as an effective Cooper pairing [16] [17–19]. Electron-phonon coupling has been studied too, but recent developments suggest that it could act as a source of additional interactions – namely, a nearest neighbours attraction mediated by phonons [20]. This idea leads to the next section, where we present and motivate the extension to the Hubbard model that we studied.

1.2 The Extended Hubbard model

The Hubbard model of Eq. (1.1) has been long considered a fair description of the main features of copper oxides [8]. There are evidences, however, that the model could be missing some interactions that must be accounted to describe d-wave superconductivity in cuprates, for instance a nearest-neighbours attraction [20]. Moreover, recent developments have shown numerically that inclusion of a non-local attraction between neighbouring sites is necessary to reproduce some features of the excitation spectrum of doped cuprates [21].

Moreover, it has been known since early Hartree-Fock calculations that adding a non-local attraction contributes to $d_{x^2-y^2}$-wave superconductivity on the square lattice [22]. With these motivation, we propose the following line of reasoning:

1. We insert "artificially" a non-local attraction in the Hubbard model, knowing that it leads to $d_{x^2-y^2}$-wave superconductivity;
2. We study the resulting model, searching for manifestations of Mott physics and understanding how the newly added interaction relates to it;
3. We analyse how antiferromagnetism and superconductivity are impacted by the findings of the step above.

To summarise: we know that Mott insulation originates $d_{x^2-y^2}$-wave superconductivity. It is worth asking if the opposite implication holds.

We define the Extended Hubbard model (EHM) by adding a non-local attraction to the Hubbard model of Eq. (1.1),

$$\hat{H}_{\text{EHM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \qquad (t, U, V > 0). \tag{1.3}$$

The newly added interaction is independent of spin index and acts on nearest neighbours. We will study the ground state of this model by the means of theoretical and numerical Hartree-Fock methods, varying the strength of the interactions U, V and electronic density. In particular, we will explore the competition of three phases (namely: normal, antiferromagnetic and superconducting) as we vary the model parameters and assess which one is the most stable. We will limit to finite square lattices and low temperatures.

1.3 Outline of this thesis

[To be updated]

In this section, we give an outline of next chapters.

Chap. **??** We introduce the Hartree-Fock method for a generic fermionic quartic hamiltonian, and discuss how it is used to find a variational approximation to its ground state. Then, we explain how this problem can be re-stated in terms of a finite set of self-consistency equations. In the end, we propose an iterative algorithmic scheme to solve these equations.

Chap. **??** We discuss the symmetries of the extended Hubbard model of Eq. (1.3). Then we give a definition of the three phases (normal, antiferromagnetic, superconducting) and describe each based on which of the original symmetries is spontaneously broken.

Chap. **??** We apply the Hartree-Fock method to the normal phase and derive the self-consistency equations. Then we present numeric results that compose evidence of V-induced Mott localisation.

Chap. **??** We apply the Hartree-Fock method to the antiferromagnetic phase and derive the self-consistency equations. First, we discuss antiferromagnetism in the conventional Hubbard model of Eq. (1.1), hence setting $V = 0$. Then we let $V > 0$ and discuss how the Hartree-Fock solution changes. In the end, we present numeric results that show how antiferromagnetic ordering is enhanced (at half-filling) by V-induced Mott localisation.

Chap. **??** We apply the Hartree-Fock method to the antiferromagnetic phase and derive the self-consistency equations. We discuss the impossibility (at Hartree-Fock level) of s-wave superconductivity in the conventional Hubbard model of Eq. (1.1), setting $V = 0$. Then we let $V > 0$ and discuss how the presence of a non-local attraction gives rise to a stable superconducting solution. In the end, we discuss numeric results that show:

a) The competition among (extended) s-wave and $d_{x^2-y^2}$-wave superconductivity, and how the two symmetries can mix in a region of parameters space;

b) The insurgence of nematic effects (namely, breaking of planar rotational symmetry) related to the mixing of gap symmetries;

c) The presence of Mott localisation in superconducting phase.

Chap. **??** We summarise our findings and *thank for all the fish*.

Appendices are organised as follows:

App. **??** We discuss the superexchange mechanism already introduced in Sec. 1.1.2.

App. **??** We define the set of gaussian states, which are used in the Hartree-Fock method discussed in Chap. **??**.

App. **??** We explain how Hartree-Fock methods and mean-field theory are related.

App. **??** We give more details about antiferromagnetism in the conventional Hubbard model.

App. **??** We derive a set of theoretical constraints on the Hartree-Fock solution for the normal and AF phase, derived respectively in Chaps. **??** and **??**.

App. **??** We discuss one concrete example of algorithmic optimisation that we used extensively for all phases. The example is discussed in the context of (absence of) s-wave superconductivity in the conventional Hubbard model.

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